

A DISENTANGLEMENT-DRIVEN DYNAMIC GRAPH LEARNING FRAMEWORK FOR INTERPRETABLE MULTIMODAL BRAIN NETWORK ANALYSIS

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ABSTRACT

Multimodal brain network analysis is critical for diagnosing neurological disorders, but existing fusion methods are limited by the entanglement of modality-shared and modality-unique information. To address this, we propose a multimodal Disentanglement Driven Graph Attention Network (DDGAT). Our model first purifies information by disentangling features into shared and unique components using dual constraints: distribution alignment and orthogonality. Subsequently, a cross-modal attention mechanism then operates exclusively on these purified shared features to dynamically construct a "neuropathology-salient" graph for each subject. This learned graph guides a Graph Attention Network to integrate the unique features in a context-aware manner, followed by an adaptive fusion module for classification. Experiments on the large-scale ADNI and ABIDE datasets demonstrate state-of-the-art diagnostic performance in classifying Alzheimer's Disease(AD) and Autism Spectrum Disorder(ASD), respectively. Furthermore, an interpretability analysis of the model-discovered biomarkers reveals aberrant connectivity patterns consistent with established neuropathology, validating the clinical plausibility of our approach.

Index Terms— Multimodal Fusion, Disentangled Representation, Dynamic Graph Learning, Brain Network Analysis, Interpretable AI

1. INTRODUCTION

Multimodal brain network analysis is a promising approach for diagnosing complex neurological disorders, from ASD [1] to AD[2]. This strategy involves integrating functional connectivity (FC) networks from fMRI with structural representations derived from diffusion tensor imaging (DTI) or advanced methods like the Morphometric Inverse Divergence (MIND) network[3]. The inherent complexity of fusing these distinct network types necessitates the use of deep learning, particularly Graph Neural Networks (GNNs), to effectively learn the intricate topological patterns associated with disease.

The primary motivation for multimodal fusion lies in leveraging the complementary information between brain function and structure. Existing methods have explored various fusion strategies, for instance, [4] use structural connectivity to enhance functional

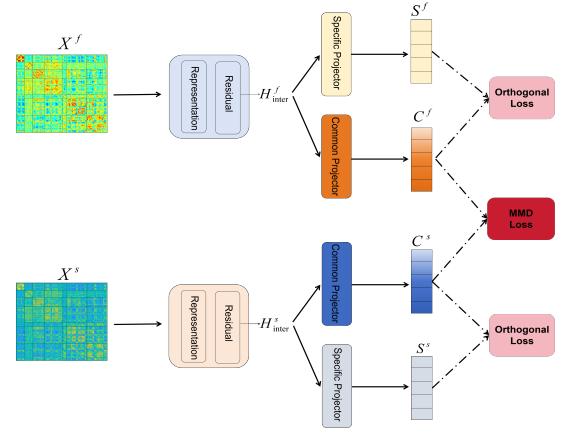


Fig. 1: Modality-Disentanglement module

graphs, while others, [5] learn a correspondence matrix to map between modalities, [6] employ attention mechanisms to integrate features. More recently, [7] have underscored the importance of learning both shared and modality-specific features, though often relying on implicit separation and pre-defined graph topologies. However these methods typically operate on entangled representations, leading to suboptimal fusion where inter-modal interference is inevitable.

To overcome these multifaceted limitations, we propose a novel and interpretable Disentanglement Driven Graph Attention Network (DDGAT) framework. As the foundational step, our model employs a feature disentanglement module [8], as illustrated in Figure1, this module utilizes independent encoders for each modality to process the brain networks (X^f , X^s). Each encoder decomposes the learned representation into common and specific components (C^m , S^m) via projection heads.

Subsequently, we introduce a cross-modal attention mechanism that operates exclusively on the common components (C^f , C^s) to dynamically learn a "neuropathology-salient fusion graph.". And then this globally informed graph serves as the backbone for a Graph Attention Network (GAT)[9] to integrate the modality-specific information (S^f , S^s) in a context-aware manner, embodying a "global-guides-local" integration principle. Finally, an adaptive attention-based fusion module provides a nuanced final decision and offers crucial modality-level interpretability.

Extensive experiments on the large-scale ABIDE [10] and ADNI [11] datasets were conducted to validate our framework. Our model achieves state-of-the-art performance compared to bench-

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mark models, and comprehensive ablation studies validate the critical contribution of each component. Finally, we visualize the top-5 aberrant brain connections identified by a Network-Based Statistic (NBS) analysis [12], demonstrating our model’s capacity for discovering clinically-relevant patterns.

2. METHOD

2.1. Overview

Our proposed framework (Fig.2) is an end-to-end network that takes a functional and a structural brain network as input. It consists of several core modules: (a) a Feature Disentanglement module separates features into common and specific components; (b) a Cross-Modal Attention module leverages only the common features to dynamically learn a fusion graph; (c) a Graph Attention Network (GAT) uses this graph to integrate specific features; (d) an Adaptive Fusion Module weighs the representations for classification.

2.2. Multimodal Feature Disentanglement

To mitigate inter-modal noise while respecting the unique characteristics of each modality, our framework employs a dual-stream feature disentanglement architecture. For each modality $m \in \{f, s\}$, an independent encoder E^m decomposes the input brain network matrix $X^m \in \mathbb{R}^{N \times N}$ into a common feature component $C^m \in \mathbb{R}^{N \times D_c}$ and a specific feature component $S^m \in \mathbb{R}^{N \times D_s}$. Each encoder is implemented as a MLP with residual connections, which first learns an intermediate representation before two independent linear projection heads produce the final common and specific components:

$$[C^m, S^m] = E^m(X^m) \quad (1)$$

This decomposition is driven by two key regularization constraints. First, a Maximum Mean Discrepancy (MMD) loss, $\mathcal{L}_{\text{mmd}}$, is applied across streams to align the distributions of common features (C^f, C^s). We adopt MMD as a discriminator-free alignment objective, which is more stable than adversarial losses in small-sample clinical settings and yields more reliable common representations in our experiments. Second, an orthogonality loss, $\mathcal{L}_{\text{ortho}}$, is applied within each stream. It minimizes the squared cosine similarity between the common and specific feature vectors (C^m, S^m), thereby encouraging them to be uncorrelated and ensuring a purer separation of information. The combined objective is:

$$\mathcal{L}_{\text{dis}} = \lambda_{\text{mmd}} \mathcal{L}_{\text{mmd}}(C^f, C^s) + \lambda_{\text{ortho}} \sum_{m \in \{f, s\}} \mathcal{L}_{\text{ortho}}(C^m, S^m) \quad (2)$$

2.3. Dynamic Graph Construction via Cross-Modal Attention

We use the disentangled common features (C^f, C^s) to construct a dynamic graph, a core innovation of our framework. This is achieved via a bi-directional cross-modal attention mechanism [13]. As showed in (Fig. 3). We compute two attention weight matrices:

$$W_{s \rightarrow f} = \text{Attention}(Q = C^f, K = C^s, V = C^s) \quad (3)$$

$$W_{f \rightarrow s} = \text{Attention}(Q = C^s, K = C^f, V = C^f) \quad (4)$$

The final dynamic adjacency matrix, A_{dyn} , is derived by symmetrically fusing and normalizing these weights:

$$A_{\text{dyn}} = \text{Softmax} \left(\frac{W_{s \rightarrow f} + W_{f \rightarrow s}^T}{2} \right) \quad (5)$$

Unlike pre-defined connectomes, A_{dyn} is learned end-to-end for each subject and task. Because the entire process is differentiable and trained end-to-end, the learned graph is optimized to accentuate the cross-modal connections most discriminative for the diagnostic task.

2.4. GAT-based Integration of Specific Information

The dynamic graph A_{dyn} provides a globally-informed context for processing the specific features (S^f, S^s) with a two-layer GAT [9]. For each modality $m \in \{f, s\}$, the features S^m serve as initial node embeddings (H_0^m). The GAT layers update them as:

$$H_1^m = \text{GATLayer}_1(H_0^m, A_{\text{dyn}}) \quad (6)$$

$$H_{\text{final}}^m = \text{GATLayer}_2(H_1^m, A_{\text{dyn}}) + H_1^m \quad (7)$$

This process embodies a key principle of our framework: using the global, consensus-based information (the dynamic graph) to guide the contextual integration of local, modality-specific details (the specific features). The GAT’s attention mechanism allows each node to adaptively weigh its neighbors as defined by A_{dyn} , effectively filtering noise and amplifying relevant signals within the specific feature space.

2.5. Adaptive Fusion and Classification

The final stage fuses the high-level representations. A global mean pooling aggregates node embeddings into graph vectors, g^f and g^s . An MLP then computes sample-specific modality importance weights, α_f and α_s :

$$[\alpha_f, \alpha_s] = \text{Softmax} \left(\text{MLP} \left([g^f; g^s] \right) \right) \quad (8)$$

The final representation, g_{final} , is a learned convex combination, which is fed to a classifier:

$$g_{\text{final}} = \alpha_f g^f + \alpha_s g^s \quad (9)$$

This adaptive weighting allows the model to dynamically rely more on one modality over the other depending on the individual case, providing a nuanced final decision. The learned weights, α_f and α_s , also offer a direct, modality-level explanation for the model’s prediction.

3. EXPERIMENTS

3.1. Datasets and Implementation

The model is evaluated on two public datasets: **ABIDE** (1,077 participants: 520 ASD, 557 NC) and **ADNI** (442 participants: 149 AD, 151 MCI, 142 NC). Data were preprocessed with standard pipelines, and brain networks were constructed using the Schaefer atlas [14], where each brain was parcellated into 100 regions. For ADNI, we perform three binary classification tasks: AD vs. NC, MCI vs. NC, and AD vs. MCI. A 5-fold cross-validation is used for evaluation, reporting the average Accuracy (Acc), AUC, Sensitivity (Sen), and Specificity (Spe) on the test folds. We compared our method against three benchmarks: M-GCN [4], Cross-GNN [5], and TAN [6]. All models were implemented in PyTorch and trained on an RTX 4070 GPU using the Adam optimizer with a batch size of 32. For the final reported results, we used a learning rate of 3×10^{-4} and a weight decay of 5×10^{-4} . We additionally report parameter counts: DDGAT has 0.63M parameters, which is smaller than Cross-GNN (**1.75M**), TAN (**1.22M**), and M-GCN (**0.85M**), indicating the improvement is not due to brute-force scaling.

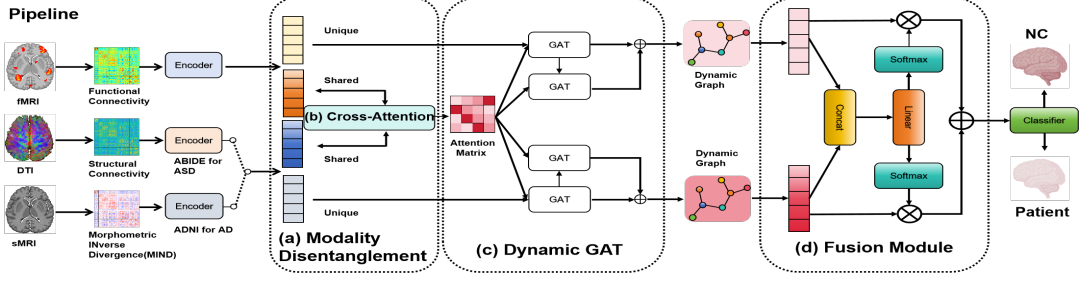


Fig. 2: The framework consists of four main modules: (a) Independent encoders decompose multimodal inputs into shared and unique features. (b) A cross-attention mechanism uses the shared features to learn a dynamic Attention Matrix. (c) This matrix serves as the topology for a GAT to integrate the unique features. (d) An adaptive attention module weighs the resulting graph representations from both streams for final classification.

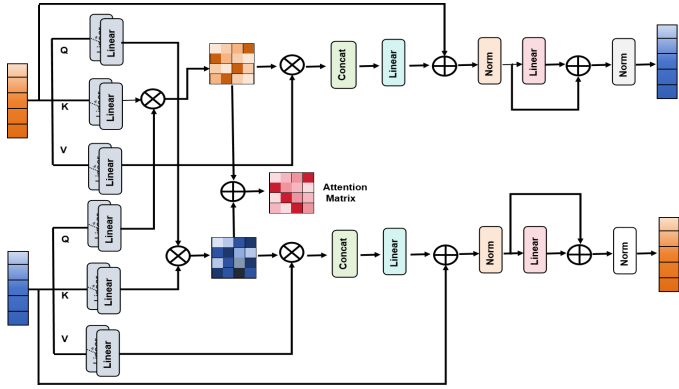


Fig. 3: The Cross-Attention module, which processes the shared features from the disentanglement stage (Fig. 2(a)). It dynamically computes a Dynamic Graph Adjacency Matrix via bi-directional attention, which is then fed to the GAT module (Fig. 2(c)).

3.2. Results and Analysis

As shown in Table 1, our proposed framework demonstrates consistent improvements across all four classification tasks, establishing a new state-of-the-art. Notably, our model achieves a substantial margin over the next best method in the ABIDE task (e.g., 4.16% higher accuracy than TAN) and shows marked superiority in the challenging early-stage ADNI classifications (e.g., MCI vs. NC and AD vs. MCI). This superior performance can be attributed to our model’s unique architecture. The foundational disentanglement module ensures that the learning process is based on purified, high-quality information, while the dynamic graph construction allows the model to discover subtle, non-linear cross-modal patterns critical for identifying nuanced disease markers.

To further validate our core mechanism, we performed a grid search with $\lambda_{mmd}, \lambda_{ortho} \in \{0, 0.25, 0.5, 0.75, 1\}$, with the results shown in Figure 4. This analysis empirically confirms the necessity of our disentanglement module, as optimal performance across all tasks is consistently achieved with non-zero values for both hyperparameters.

3.3. Ablation Study

Table 2 presents the results of our ablation study, which systematically validates the contribution of each key component. Removing

Table 1: Performance comparison with benchmark models (mean $\pm$ std %). Metrics shown are Acc, AUC, Sen, and Spe. Best: **bold**; second-best: underlined. The term * indicates statistically significant improvements ($p < 0.05$)

Task	Method	Acc	AUC	Sen	Spe
ABIDE (ASD/NC)	M-GCN [4]	65.64 $\pm$ 2.14	64.96 $\pm$ 2.42	62.53 $\pm$ 16.96	68.50 $\pm$ 19.44
	Cross-GNN [5]	64.57 $\pm$ 3.02	64.21 $\pm$ 3.17	63.97 $\pm$ 5.46	65.19 $\pm$ 9.24
	TAN [6]	<u>66.80$\pm$1.78</u>	<u>68.46$\pm$2.93</u>	59.35 $\pm$ 0.88	73.78$\pm$3.71
	Ours	70.96$\pm$1.63*	73.51$\pm$0.59	68.49$\pm$3.12	73.26$\pm$0.20
ADNI (AD/NC)	M-GCN	72.83 $\pm$ 6.83	72.19 $\pm$ 7.38	78.57 $\pm$ 9.26	66.75 $\pm$ 16.25
	Cross-GNN	<u>81.09$\pm$3.33</u>	<u>84.80$\pm$3.54</u>	83.26$\pm$7.54	<u>78.87$\pm$12.25</u>
	TAN	76.62 $\pm$ 5.01	80.36 $\pm$ 7.37	75.86 $\pm$ 7.94	77.44 $\pm$ 4.94
	Ours	82.10$\pm$8.38*	85.42$\pm$9.30	<u>81.84$\pm$8.17</u>	82.34$\pm$10.02
ADNI (MCI/NC)	M-GCN	65.22 $\pm$ 4.91	64.55 $\pm$ 6.11	72.11 $\pm$ 8.73	57.64 $\pm$ 16.63
	Cross-GNN	66.90 $\pm$ 6.32	65.32 $\pm$ 2.85	66.88 $\pm$ 15.06	56.40 $\pm$ 8.18
	TAN	<u>67.24$\pm$2.73</u>	<u>65.80$\pm$2.09</u>	<u>60.57$\pm$4.07</u>	<u>60.57$\pm$4.07</u>
	Ours	68.70$\pm$2.10*	68.89$\pm$8.48	76.77$\pm$8.78	70.59$\pm$16.00
ADNI (AD/MCI)	M-GCN	62.33 $\pm$ 4.55	61.21 $\pm$ 6.41	65.66 $\pm$ 4.92	68.77 $\pm$ 10.82
	Cross-GNN	<u>68.00$\pm$3.86</u>	<u>66.46$\pm$3.33</u>	60.46 $\pm$ 9.79	<u>69.48$\pm$6.30</u>
	TAN	65.67 $\pm$ 4.03	64.45 $\pm$ 8.85	63.20 $\pm$ 9.66	68.24 $\pm$ 8.73
	Ours	70.33$\pm$5.31*	73.62$\pm$6.37	66.34$\pm$9.20	70.11$\pm$5.65

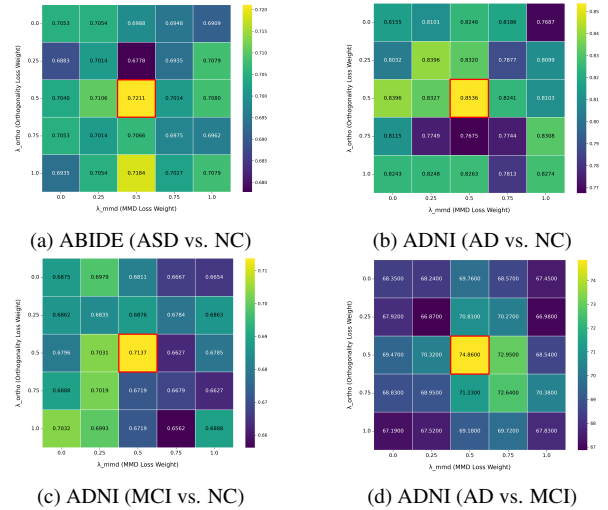


Fig. 4: Parameter sensitivity analysis of λ_{mmd} and λ_{ortho} . Colors show accuracy.

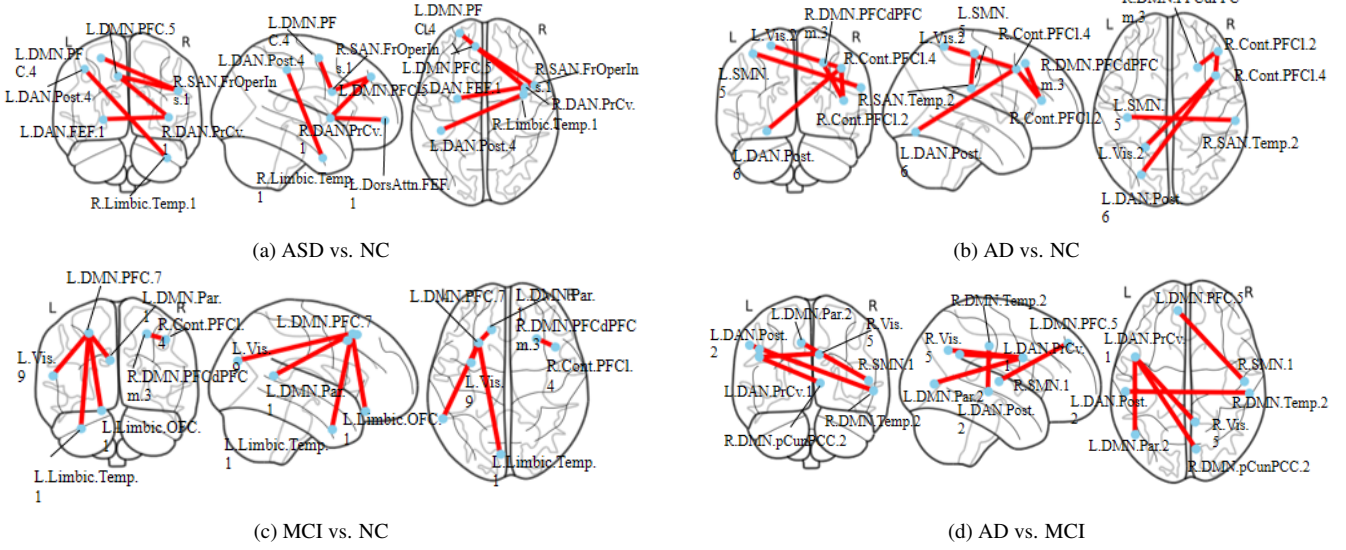


Fig. 5: Top significant aberrant connections identified by NBS analysis

the Modality Disentanglement module (a) leads to a consistent and significant drop in performance (e.g., 3.77% in ADNI AD/NC Acc), empirically confirming our hypothesis that mitigating inter-modal noise is critical for robust fusion. Moreover, the term-level ablation in Figure 4 shows that setting λ_{mmd} or λ_{ortho} to 0 reduces accuracy (e.g., on ABIDE by 3.24% and 1.76%, respectively). This confirms that both alignment and orthogonality are necessary. Notable performance degradations are observed when removing the Cross-Modal Attention mechanism for dynamic graph construction (b) or the GAT integration (c), highlighting the importance of learning an adaptive, task-aware topology and performing context-aware message passing. Although removing GAT yields a larger drop in some tasks, this reflects that message passing is the primary integrator, while disentanglement improves the graph/embeddings that GAT aggregates, indicating a strong “purify-then-aggregate” coupling. The removal of Adaptive Fusion (d) also results in noticeable performance drops, underscoring the necessity of intelligent final-stage fusion.

Table 2: Ablation study results (ACC %) across four tasks.

Variant	Task			
	ABIDE ASD/NC	ADNI AD/NC	ADNI MCI/NC	ADNI AD/MCI
Full Model	70.96	82.10	68.70	70.33
w/o Disent. (a)	67.72	78.33	65.18	66.00
w/o X-Attn (b)	67.19	78.33	64.16	64.67
w/o GAT (c)	64.31	76.61	65.20	66.00
w/o Fusion (d)	64.83	77.32	63.83	67.33

3.4. Interpretability Analysis

To identify key brain connectivities underlying the model’s decisions, we applied NBS analysis [12]. This method controls for false positives in whole-brain connectomic studies by performing statistical inference at the subnetwork level, rather than individual connections, thereby enhancing reliability in detecting disease-related patterns. The resulting analysis offers a transparent view

into the model’s decision-making process, revealing clinically relevant biomarkers as illustrated in Fig. 5. For ASD, the model identified aberrant connections between the Default Mode Network (DMN) and Salience Network (SN), localizing to core hubs like the prefrontal cortex and insula, consistent with prior findings [15]. Furthermore, the identified pattern of long-range hypoconnectivity posits the social cognition deficits in ASD are linked to reduced long-distance neural communication[16]. In the AD spectrum, our model captured the well-documented decoupling between the Executive Control Network (ECN) and DMN [17]. For MCI, the identified weakened DMN-ECN links reflect a breakdown in the brain’s modular architecture—a phenomenon often described as reduced network segregation [18] or dedifferentiation [19]—which is a state known to correlate with amyloid pathology [20]. Critically, for the AD vs. MCI task, the model highlighted pathological hyperconnectivity between the DMN and dorsal attention network, a pattern indicative of a failure to separate task-negative and task-positive states.[21].

4. CONCLUSION

In this paper, we proposed a novel and interpretable multimodal graph learning framework that addresses critical challenges in brain network analysis. Our key innovation is a disentanglement-driven architecture that introduces a principled approach to fusion. By first purifying information into shared and specific components, our model leverages these consensus-based features to dynamically construct a task-aware graph. This learned topology then guides the contextual integration of modality-specific information, representing a reorganization from static connectomes. Our framework achieves state-of-the-art classification performance on the large-scale ADNI and ABIDE datasets. Crucially, this high accuracy is complemented by powerful interpretability, as an NBS analysis identified clinically-relevant aberrant connectivity patterns, such as long-range hypoconnectivity, consistent with established neuropathology. This work presents a promising step for neurological disorder diagnosis.

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